Scaling, Sensitivity, and Structure: The N-Dependent Geometry of ED-Arch Collapse Surfaces

Allen Proxmire
March 2026

Abstract
Collapse dynamics in Event Density Architecture (ED-Arch) depend fundamentally on the number of particles N in the ring. Laws I–III established the timing, resonance, and recurrence geometry of the tangent gate for a fixed N, but empirical sweeps reveal that these structures transform systematically as N increases. Law IV derives the scaling laws that govern this transformation. The effective circumference grows linearly with N, causing drift-phase velocity to increase, recurrence time to shrink, and recurrence windows to narrow. As a result, ladder spacing scales as $\Delta(N) \propto 1/N$, DECAY spacing scales as $\Delta d(N) \propto 1/N$, and the density of both ladder rungs and DECAY islands grows proportionally to N.

These scaling laws explain why collapse surfaces become sharper, denser, and more corrugated for larger rings. Recurrence windows and resonance islands shrink together, producing increasingly intricate interleaving and a more finely serrated tangent-gate ridgeline. When collapse surfaces are rescaled by N, they converge to universal curves, revealing that ED-Arch collapse is governed by self-similar geometric families across particle count.

Law IV establishes N as the master parameter of ED-Arch. It provides the scaling framework that unifies timing, resonance, and recurrence, and sets the stage for Law V, which will classify the commensurability structures that govern alignment and misalignment across all N.

1. Introduction
Event Density Architecture (ED-Arch) reveals that collapse dynamics in periodic geometries are governed by discrete geometric structures. Laws I–III established the timing, resonance, and recurrence geometry of the tangent gate for a fixed number of particles N. These laws explain how collapse occurs: through quantized recurrence windows, rational resonance, and the ladder of collapse times.

But ED-Arch is fundamentally an N-dependent system.
As the number of particles increases, the geometry of the ring changes, the drift-phase velocity increases, recurrence windows narrow, resonance spacing shrinks, and the tangent gate becomes increasingly intricate. The collapse surfaces for N=3 are only the first instance of a broader scaling structure.

Law IV explains how collapse surfaces scale with N.

It shows that:

- drift-phase velocity grows linearly with N
- recurrence windows shrink like $1/N$
- DECAY spacing shrinks like $1/N$
- ladder density grows like N
- collapse surfaces become sharper, denser, and more corrugated
- the tangent gate becomes more resonant as N increases

In other words:

The complexity of the tangent gate is an increasing function of *N*.

Law IV provides the first unified scaling law for ED-Arch collapse dynamics. It generalizes the timing, resonance, and recurrence structures of Laws I–III across all particle counts and reveals the universal patterns that emerge when collapse surfaces are rescaled by *N*.

1.1 The Need for an N-Scaling Law

Laws I–III describe the geometry of collapse for a fixed *N*.
But empirical sweeps show that:

- ladder rungs multiply as *N* increases
- DECAY islands become more frequent
- resonance and recurrence interleave more densely
- collapse surfaces sharpen and fragment
- mechanism boundaries shift systematically

These changes are not incidental.
They follow a predictable geometric pattern.

Law IV explains this pattern by deriving how the fundamental quantities of ED-Arch scale with *N*:

- circumference
- drift-phase velocity
- recurrence time
- resonance spacing
- window width
- collapse sensitivity

This scaling law is essential for understanding ED-Arch as a unified system rather than a collection of isolated cases.

1.2 Purpose of Law IV

Law IV provides the first analytic derivation of:

- how recurrence time $T_r(N)$ scales
- how ladder spacing $\Delta(N)$ scales
- how DECAY spacing $\Delta d(N)$ scales
- how recurrence window width $W(N)$ scales
- how drift-phase velocity $\dot{\phi}(N)$ scales
- how collapse surfaces transform under *N*-rescaling

It explains:

- why larger rings produce more intricate tangent gates
- why collapse becomes more sensitive as *N* increases

- why resonance and recurrence become denser
- why collapse surfaces converge to universal shapes when rescaled

Law IV is the scaling backbone of ED-Arch.

1.3 Relationship to Laws I–III

- Law I: quantized timing
- Law II: rational resonance
- Law III: discrete recurrence
- Law IV: scaling across N

Law IV generalizes the first three laws by showing how their structures evolve as the number of particles increases. It reveals that timing, resonance, and recurrence are not fixed geometric objects—they are scaling families.

Together, Laws I–IV form the geometric, resonant, temporal, and scaling architecture of ED-Arch collapse.

2. Geometry of N-Scaling

The geometry of a ring of N particles determines the drift-phase velocity, recurrence time, resonance spacing, and collapse sensitivity. Law IV begins by deriving how the fundamental geometric quantities scale with N.

2.1 Circumference Scaling

For a ring of N particles of diameter d, the effective circumference is:

$$C(d,N) = Nd\,\alpha(N),$$

where $\alpha(N)$ is a geometric factor determined by the polygonal spacing of the particles.

To first order:

$$C(d,N) \propto N.$$

This linear scaling is the foundation of all subsequent scaling laws.

2.2 Drift-Phase Velocity Scaling

The drift-phase velocity is:

$$\dot{\phi}(N) = (C(d,N) / L) \cdot 2\pi.$$

Thus:

$$\dot{\phi}(N) \propto N.$$

As N increases:

- the ring drifts faster

- phase accumulates more rapidly
- recurrence windows narrow
- resonance becomes denser

This is the geometric origin of the increasing complexity of the tangent gate.

2.3 Recurrence Time Scaling
The recurrence time is:

$$T_r(N) = L / v_{\text{drift}}(N).$$

Because drift speed increases with circumference:

$$T_r(N) \propto 1 / N.$$

Thus:

- larger $N \rightarrow$ shorter recurrence time
- shorter recurrence time $\rightarrow$ more ladder rungs
- more rungs $\rightarrow$ finer quantization

This explains why the ladder becomes denser as N increases.

2.4 Collapse Sensitivity Scaling
Collapse sensitivity is governed by the width of recurrence windows:

$$W(N) \propto 1 / \dot{\phi}(N) \propto 1 / N.$$

Thus:

- larger $N \rightarrow$ narrower windows
- narrower windows $\rightarrow$ greater sensitivity
- greater sensitivity $\rightarrow$ more DECAY islands

This matches the empirical behavior of the tangent gate.

2.5 Why Larger N Produces More Structure
Combining the above:

- drift-phase velocity grows like N
- recurrence time shrinks like $1/N$
- window width shrinks like $1/N$
- DECAY spacing shrinks like $1/N$
- ladder density grows like N

Thus:

The tangent gate becomes more resonant, more quantized, and more intricate as N increases.

This is the central insight of Law IV.

3. Scaling of Recurrence Windows

Recurrence windows are the temporal intervals during which collapse is geometrically possible. Law III showed that these windows occur at integer multiples of the recurrence time T_r. Law IV now explains how the width, density, and structure of these windows scale with the number of particles N.

As N increases:

- drift-phase velocity increases
- recurrence time decreases
- window width shrinks
- windows become more densely packed
- collapse becomes more sensitive

This is the first indication that the tangent gate becomes more intricate as N grows.

3.1 Window Width Shrinks Like 1/N

The width of a recurrence window is:

$$W(N) \approx \epsilon / \dot{\phi}(N).$$

Since:

$$\dot{\phi}(N) \propto N,$$

we obtain the fundamental scaling law:

$$W(N) \propto 1 / N.$$

Thus:

- larger $N \rightarrow$ narrower windows
- narrower windows $\rightarrow$ greater sensitivity
- greater sensitivity $\rightarrow$ more DECAY islands

This matches the empirical behavior of the tangent gate across all tested N.

3.2 Window Density Grows Like N

The recurrence time is:

$$T_r(N) \propto 1 / N.$$

Thus, the number of recurrence windows per unit time is:

$$\text{density}(N) \propto N.$$

As N increases:

- recurrence windows become more frequent
- collapse opportunities multiply
- the ladder becomes more densely populated

This explains why the tangent gate becomes more finely quantized for larger rings.

3.3 Interaction with Drift Phase

The drift phase evolves as:

$$\phi(t) = \dot{\phi}(N)\, t.$$

Because $\dot{\phi}(N)$ grows with N:

- phase accumulates faster
- alignment becomes more sensitive
- misalignment becomes more common

This is the geometric origin of the increasing complexity of the tangent gate.

3.4 Consequences for Collapse

Combining the above:

- more windows
- narrower windows
- faster phase accumulation

produces:

- sharper transitions between rungs
- more DECAY islands
- more intricate ridgeline geometry
- greater sensitivity to particle diameter d

The tangent gate becomes more structured as N increases.

4. Scaling of the Ladder

The ladder of collapse times — first identified in Law I — is the temporal imprint of recurrence geometry. Law IV now shows how the ladder itself scales with N. As the number of particles increases, the ladder becomes denser, finer, and more sensitive, reflecting the underlying scaling of recurrence windows.

4.1 Ladder Step Size Shrinks Like $1/N$

Law I observed:

$$\chi = \chi_0 + k\Delta.$$

Law III identified:

$$\Delta = T_r.$$

Law IV now shows:

$$T_r(N) \propto 1 / N.$$

Thus:

$$\Delta(N) \propto 1 / N.$$

This is the fundamental scaling law of the ladder:

- larger $N \rightarrow$ smaller step size
- smaller step size $\rightarrow$ more rungs
- more rungs $\rightarrow$ finer quantization

The ladder becomes denser as N increases.

4.2 Ladder Density Grows Like N

Because the step size shrinks like $1/N$, the number of rungs per unit time grows like:

$$\text{density}(N) \propto N.$$

This explains:

- why collapse surfaces become more corrugated
- why the tangent gate becomes more structured
- why mechanism boundaries become sharper

The ladder is not fixed — it is a scaling family.

4.3 Why the Ladder Becomes Finer

The ladder becomes finer because:

- drift speed increases
- recurrence time decreases
- windows become more frequent
- collapse opportunities multiply

This produces a ladder with:

- more rungs
- tighter spacing
- sharper transitions

The tangent gate becomes more quantized as N increases.

4.4 Universality Across N

The ladder exists for all N, but its density and structure scale predictably:

- step size $\Delta(N) \propto 1/N$
- rung density $\propto N$
- window width $\propto 1/N$
- sensitivity $\propto N$

Thus:

> **The ladder is a universal structure whose geometry is controlled by N.**

This universality is the foundational observation of Law IV.

5. Scaling of DECAY Resonance

Law II established that DECAY arises from rational misalignment between ring circumference and box geometry. Law IV now shows how the density, spacing, and strength of these resonances scale with the number of particles N. As N increases, the resonance structure becomes denser, narrower, and more intricate, producing the increasingly serrated geometry of the tangent gate.

The central result is simple and powerful:

> **DECAY spacing shrinks like $1/N$, and DECAY density grows like N.**

This scaling law explains why the tangent gate becomes more resonant as the ring grows.

5.1 DECAY Spacing Shrinks Like $1/N$

Law II showed that DECAY islands occur when:

$$C(d,N) / L \approx p / q.$$

Since:

$$C(d,N) \propto N,$$

the spacing between adjacent rational approximants in d scales as:

$$\Delta d(N) \propto 1 / N.$$

Thus:

- larger $N \to$ smaller DECAY spacing
- smaller spacing $\to$ more DECAY islands
- more islands $\to$ more serrated collapse surfaces

This matches empirical sweeps across all tested N.

5.2 Resonance Density Grows Like N
Because DECAY spacing shrinks like $1/N$, the number of DECAY islands per unit interval of d grows like:

$$\text{density}(N) \propto N.$$

This explains:

- why the tangent gate becomes more densely resonant
- why collapse surfaces become more corrugated
- why mechanism boundaries become sharper

Resonance is not fixed — it is a scaling family.

5.3 Interaction with Recurrence
Law III showed that recurrence windows shrink like $1/N$.
Law IV now shows that DECAY spacing shrinks at the same rate:

$$W(N) \propto \Delta d(N) \propto 1 / N.$$

Thus:

- recurrence windows and resonance islands shrink together
- their interleaving becomes denser
- the broken ridgeline becomes more intricate

This is the geometric origin of the increasing complexity of the tangent gate.

5.4 Consequences for Collapse
Combining the above:

- more resonances
- narrower resonances
- more recurrence windows
- narrower windows

produces:

- more DECAY islands
- more ladder rungs
- more complex ridgeline geometry

- greater sensitivity to particle diameter *d*

The tangent gate becomes more resonant as *N*increases.

6. Scaling of Collapse Surfaces

Collapse surfaces are the global geometric objects that encode collapse time as a function of particle diameter *d* and initial phase. Law IV now shows how these surfaces transform as *N* increases. The result is a unified scaling law that explains why collapse surfaces become sharper, denser, and more corrugated for larger rings.

The structural insight is:

> **Collapse surfaces are not fixed objects — they are scaling families controlled by *N*.**

6.1 Collapse Surface Geometry Sharpens with N

As *N* increases:

- recurrence windows narrow
- resonance islands narrow
- ladder rungs multiply
- mechanism boundaries sharpen

This produces collapse surfaces with:

- steeper gradients
- sharper transitions
- more intricate structure

The tangent gate becomes more geometrically expressive as *N* grows.

6.2 N-Dependent Mechanism Boundaries

The boundaries between:

- inward sheet
- outward mirror
- tangent ladder
- DECAY islands
- shift systematically with *N*.

Because:

- drift speed increases
- recurrence time decreases
- resonance density increases

the mechanism boundaries become:

- sharper
- more numerous
- more sensitive

This explains the increasingly complex mechanism maps observed in high-N sweeps.

6.3 Scaling of Collapse Time
Law I gave:

$$\chi = \chi_0 + k\Delta.$$

Law III identified:

$$\Delta = T_r.$$

Law IV now shows:

$$T_r(N) \propto 1 / N.$$

Thus:

$$\chi(N) = \chi_0 + k(c / N),$$

for some constant c determined by drift speed and box geometry.

This predicts:

- shorter collapse times for larger N
- more rungs in the same time interval
- finer temporal quantization

Collapse time itself becomes an N-scaled quantity.

6.4 Emergence of Universal Curves
When collapse surfaces are rescaled by N:

- ladder spacing becomes constant
- DECAY spacing becomes constant
- recurrence windows become constant
- mechanism boundaries align

This reveals that collapse surfaces converge to universal shapes under N-rescaling.

In other words:

ED-Arch collapse surfaces are self-similar across N.

This is the organizing principle of Law IV.

7. Discussion

Law IV reveals that the complexity of the tangent gate is not an incidental feature of ED-Arch — it is a scaling law. As the number of particles N increases, the geometry of the ring changes in ways that systematically reshape the timing, resonance, and recurrence structures established in Laws I–III. The result is a predictable increase in the density, sharpness, and intricacy of collapse surfaces.

7.1 N as the Master Parameter

The number of particles N controls the fundamental geometric quantities of ED-Arch:

- circumference $C(d,N)$
- drift-phase velocity $\dot{\phi}(N)$
- recurrence time $Tr(N)$
- window width $W(N)$
- DECAY spacing $\Delta d(N)$

Because each of these scales linearly or inversely with N, the entire collapse landscape transforms in a predictable way. N is not a secondary parameter — it is the master parameter that governs the structure of the tangent gate.

7.2 Why Larger N Produces More Complexity

As N increases:

- drift speed increases
- recurrence windows shrink
- resonance spacing shrinks
- ladder rungs multiply
- DECAY islands multiply
- mechanism boundaries sharpen

These effects compound, producing collapse surfaces that are:

- more corrugated
- more sensitive
- more densely structured
- more resonant

The tangent gate becomes richer, not noisier, as N grows.

7.3 Relationship to Laws I–III

Law IV generalizes the first three laws:

- Law I: ladder spacing becomes $\Delta(N) \propto 1/N$
- Law II: DECAY spacing becomes $\Delta d(N) \propto 1/N$
- Law III: recurrence windows shrink like $W(N) \propto 1/N$

Thus:

- timing becomes finer
- resonance becomes denser
- recurrence becomes sharper

Law IV shows that these structures are not fixed — they are scaling families.

7.4 Predictability and Universality
The scaling laws derived in Law IV allow prediction of:

- collapse times
- mechanism boundaries
- resonance density
- recurrence structure
- collapse-surface geometry

for any *N*.

When collapse surfaces are rescaled by *N*, they converge to universal curves. This universality elevates ED-Arch from an empirical architecture to a predictive mathematical system.

7.5 Law IV as the Bridge to Law V
Law IV sets the stage for the final law of the series:

- Law V: Commensurability Classes

Law V will classify the rational structures that govern alignment and misalignment across all *N*. Law IV provides the scaling framework that Law V will build upon.

8. Future Work
Law IV opens several natural directions for further study. These directions extend the scaling framework, generalize it across larger *N*, and integrate it with the resonance and recurrence structures of Laws I–III.

8.1 High-Resolution N-Sweeps
Future work should map collapse surfaces for:

- N=3 (baseline)
- N=4,5,6,7,8 (scaling regime)

These sweeps will reveal:

- how ladder density grows
- how DECAY density grows
- how mechanism boundaries shift
- how collapse surfaces sharpen

Such maps will test the predictions of Law IV.

8.2 Analytic Drift-Speed Model
The recurrence time:

$$T_r(N) = L / v_{\text{drift}}(N)$$

depends on the tangential drift speed, which is currently measured empirically.
A closed-form expression for $v_{\text{drift}}(N)$ would allow:

- exact prediction of ladder spacing
- analytic classification of recurrence windows
- extension of Law IV to arbitrary N

This is a key step toward a unified analytic theory.

8.3 Universal Scaling Curves
Collapse surfaces should be rescaled by:

- N (for time)
- $1/N$ (for diameter spacing)

to reveal universal curves.
Identifying these curves will unify collapse surfaces across all N.

8.4 Recurrence–Resonance Scaling Diagrams
Plotting recurrence windows and resonance islands as functions of N will reveal:

- where constructive alignment persists
- where destructive alignment dominates
- how the broken ridgeline evolves
- how resonance strength varies with N

These diagrams will unify Laws II–IV.

8.5 Toward Law V: Commensurability Classes
The final law of the ED-Arch series will classify:

- which d values produce alignment
- which produce misalignment
- which produce long-term drift
- which produce stable periodicity

Law IV provides the scaling framework that Law V will complete.